

Department of Chemical Engineering
Chemical Reaction Engineering II (CL24303)

MO2026
 Tutorial 4

Date Assigned	25.09.2026	Submission Date and Time	04.10.2026 23:59 h
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1. A nickel catalyst for the hydrogenation of ethylene has a mean pore radius (r) of 50 Å.
- (a) Calculate the bulk and Knudsen diffusivities of hydrogen for this catalyst at 100 °C, and 1 and 10 atm pressures, in a hydrogen-ethene mixture.

$$\mathcal{D}_{C_2H_4-H_2} = \frac{0.86}{p_t} [\text{cm}^2/\text{s}]$$

$$\mathcal{D}_{Knudsen-H_2} = 9.70 \times 10^3 r \left(\frac{T}{M_{H_2}} \right)^{1/2} [\text{cm}^2/\text{s}]$$

- (b) Calculate the combined diffusivity of hydrogen

$$\frac{1}{\mathcal{D}_{H_2}} = \frac{1}{\mathcal{D}_{C_2H_4-H_2}} + \frac{1}{\mathcal{D}_{Knudsen-H_2}}$$

- (c) Calculate the effective diffusivity of hydrogen ($\mathcal{D}_e$) for the catalyst pellet if tortuosity (τ) = 4, porosity (ε) = 0.4, and constriction factor (σ) = 0.8.

$$\mathcal{D}_e = \frac{\mathcal{D}_{H_2} \varepsilon \sigma}{\tau}$$

2. A second-order reaction $A \rightarrow R$ is carried out with a catalyst of diameter 0.5 mm and pellet density of 2.2 g/cm³. A rate of 10⁵ mole/hr kg-cat is observed for a concentration of A of 20 mol/m³. If $D_e = 5 \times 10^{-5} \text{ m}^2 \text{ h}^{-1}$, is there a possibility of strong pore diffusion resistance?
3. The first-order reaction $A \rightarrow B$ was carried out using catalyst pellets of two sizes. Which of the following runs were obtained under strong pore diffusion?
 Note: pellet density of 1.6 g/cm³ and $D_e = 2 \times 10^{-6} \text{ m}^2/\text{s}$,

	$-r_{obs}' (\text{mol/g cat s}) \times 10^5$	Pellet Radius (m)
Run 1	3.0	0.01
Run 2	15.0	0.001

4. A first-order, heterogeneous, irreversible reaction is taking place within a spherical catalyst pellet that is plated with platinum throughout. The concentration at the external surface is 0.001 g-mol/dm³ (C_{AS}), the diameter ($2R$) is 2x10⁻¹cm, and the effective diffusion coefficient (D_e) is 0.1cm²/s. The concentration profile inside a spherical catalyst pellet in the presence of pore diffusion and first-order reaction is given by:

$$\frac{C_A}{C_{AS}} = \frac{1}{\xi} \left(\frac{\sinh \xi \phi}{\sinh \phi} \right) \text{ Here, } \xi = r/R, \phi = \frac{R}{3} \sqrt{\frac{k}{D_e}}$$

- (a) What is the concentration of reactant at 3x10⁻²cm from the external pellet surface?
 (b) To what diameter should the pellet be reduced if the effectiveness factor is to be 0.8?

5. An experimental rate measurement on the first-order decomposition of A is made with a particular catalyst. The following details are known:

For the spherical particle:

$$d_p = 2.4 \text{ mm}$$

$$D_e = 5 \times 10^{-5} \text{ m}^2/\text{hr} \text{ (effective mass conductivity)}$$

$$k_g = 300 \text{ m}^3/\text{hr m}^2 \text{ cat (mass transfer coefficient)}$$

$$\text{Catalyst density } (\rho) = 2.8 \text{ g/cm}^3$$

$$\text{Porosity } (\varepsilon_p) = 0.4$$

$$C_{Ag} = 20 \text{ mol/m}^3 \text{ (at 1 atm and 336}^\circ\text{C)}$$

$$-r'''_{\text{obs}} = 10^5 \text{ mol/hr. m}^3 \text{ cat}$$

Using Weisz-Prater and Mears' criterion, determine if the observed rate is limited by internal pore diffusion, external mass transfer or both?

6. A packed-bed reactor converts A to R by a first-order catalytic reaction, $A \rightarrow R$. With 9-mm pellets, the reactor operates in the strong pore diffusion resistance regime and gives 63.2% conversion. If these pellets were replaced by 18-mm pellets (to reduce pressure drop), how would this affect the conversion?

7. Write a Matlab/Python Code to determine the concentration gradient inside a porous flat catalyst in the presence of diffusion and reaction for n^{th} order reaction of $A \rightarrow B$, $-r_A = kC_A^n$

$$\frac{d^2 \tilde{C}_A}{d\xi^2} - \phi^2 \tilde{C}_A^n = 0$$

When subject to the following boundary conditions:

$$@ \xi=1 \quad \tilde{C}_A = 1$$

$$@ \xi=0 \quad \frac{d\tilde{C}_A}{d\xi} = 0$$

- (a) The output of the code should be a plot of $\tilde{C}_A$ vs ξ for $\phi = 0.1, 1, 10, 100$. Also, in the same figure, plot for $n=1$. The analytical solution for $n=1$ is given by:

$$\tilde{C}_A = \frac{\cosh(\xi\phi)}{\cosh\phi}$$

- (b) Further, plot $\tilde{C}_A$ vs ξ for $\phi = 0.1, 100$ and $n = 0.5, 2$ and 1.5 . Thereafter, interpret the trends in your own words. However, screenshots are not allowed

8. The Thiele modulus (ϕ) and the internal effectiveness factor (η) for 1st order reaction with rate constant k in catalyst particles of different shapes are given below:

Shape	ϕ	η
Flat Plate	$L \sqrt{\frac{k}{D_e}}$	$\frac{\tanh\phi}{\phi}$
Sphere	$\frac{R}{3} \sqrt{\frac{k}{D_e}}$	$\frac{1}{3\phi^2} (3\phi \coth 3\phi - 1)$
Cylinder	$\frac{R}{2} \sqrt{\frac{k}{D_e}}$	$\frac{1}{\phi} \frac{I_1(2\phi)}{I_0(2\phi)}$

Write a code in MATLAB/PYTHON to calculate η for $\phi = 1-100$ for all the shapes. Thereafter, generate a plot of η vs ϕ and comment on the physical significance of the trends. A sample plot generated in MATLAB is shown below.

